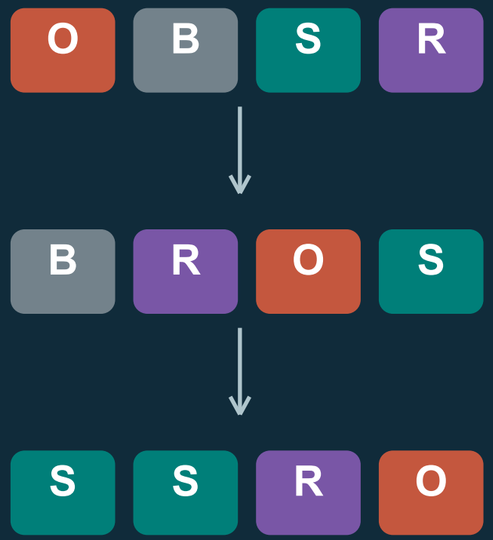
EMNLP 2026 · ORAL

DiffIE

Diffusion-based Open Information Extraction

Stochastic trajectories. Multiple valid facts.



ROLE-TAG DIFFUSION

Konstantin Fedorov · Valentin Malykh

ISDCT SB RAS · ITMO University · MWS AI
Trusted AI Research Center, RAS · IITU University

One sentence. More than one correct extraction.

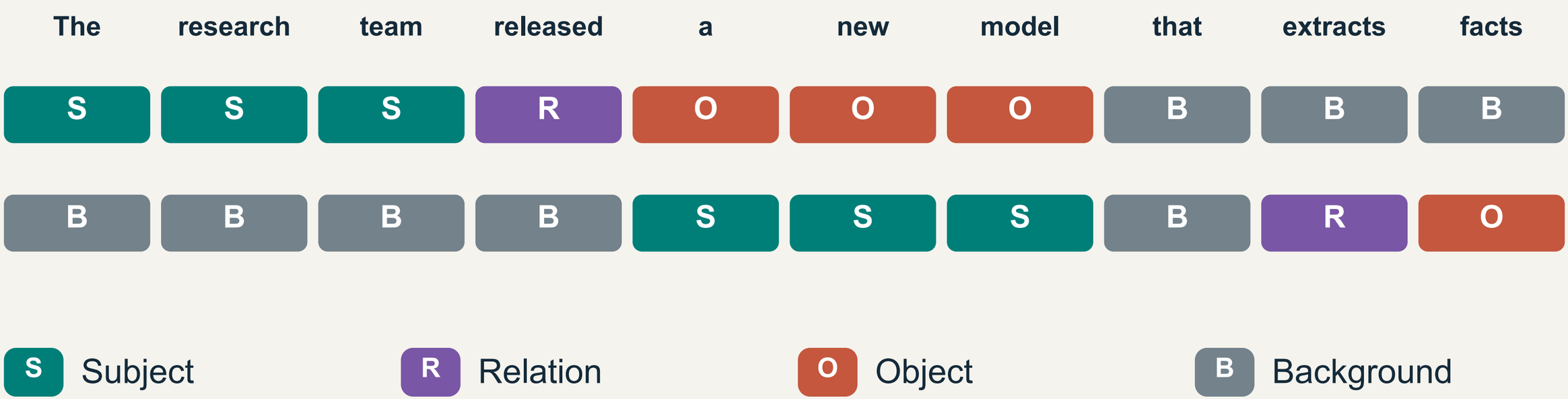
> **“The research team released a new model that extracts facts.”**

TWO VALID FACTS

The research team	released	a new model
a new model	extracts	facts

Learn a distribution over valid extractions; sample to recover the set.

Diffuse four role tags, conditioned on the sentence.



One trajectory → one role-tag sequence → at most one triplet

Paper §3.1 · B = Background (not a BIO boundary tag) · Example is schematic

Corrupt a gold tagging. Learn to recover it.



$$q(y_i^{(t)} \mid y_i^{(0)}) = \bar{\alpha}_t \cdot \text{one-hot}(y_i^{(0)}) + (1 - \bar{\alpha}_t) \cdot \text{Uniform}(4)$$

SAMPLE A TIMESTEP

t sampled uniformly

CONDITION ON THE INPUT

Sentence + noisy tags + t

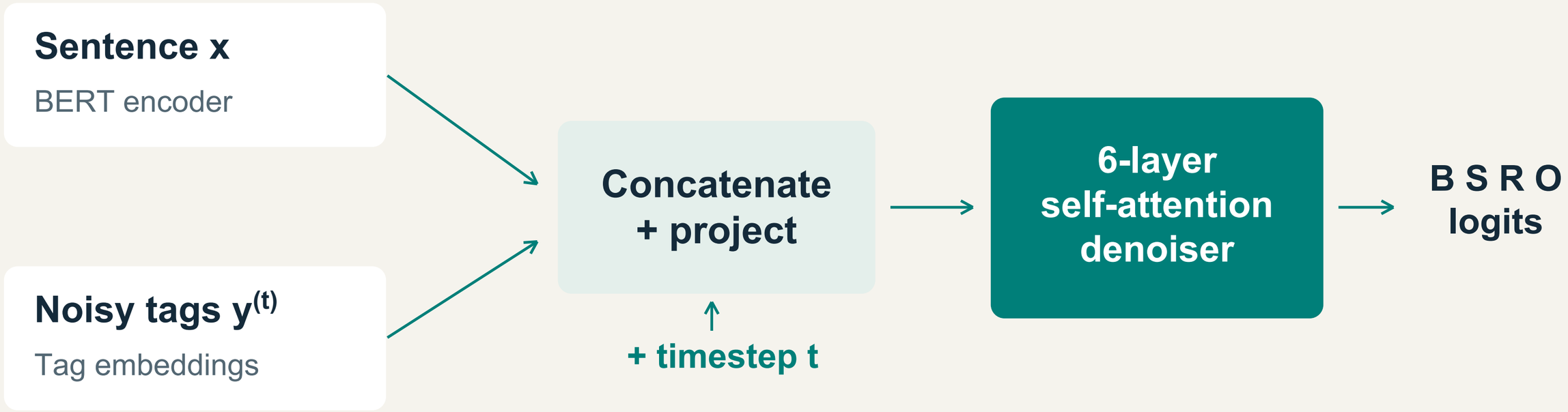
SUPERVISE CLEAN TAGS

Weighted cross-entropy

m gold triplets → m training pairs for the same sentence

Paper §3.3, Eq. (1) · Uniform discrete diffusion (D3PM; Austin et al., 2021)

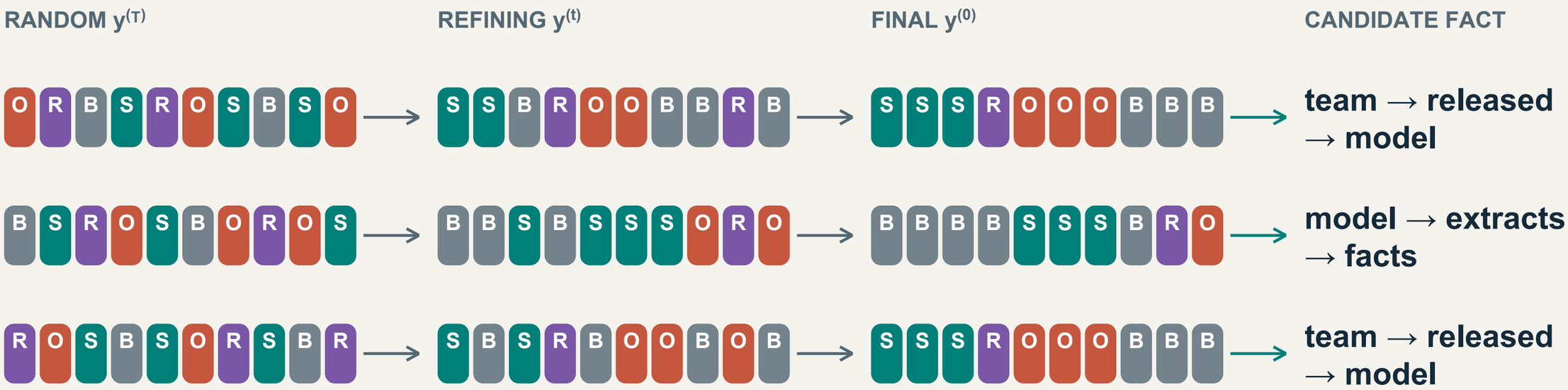
Encode once. Reuse context at every denoising step.



Shared sentence context + evolving tag state → jointly refined role predictions

Use diffusion stochasticity as the extraction mechanism.

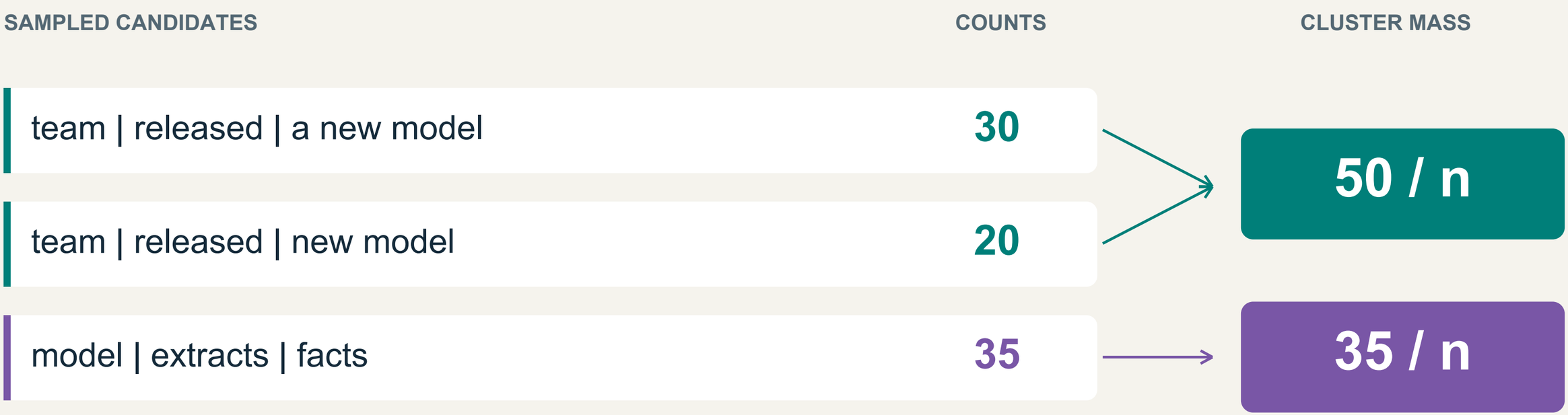
SAME ENCODED SENTENCE



n trajectories in parallel 16 sequential denoising steps per trajectory

Increase the candidate pool after training by changing n.

Aggregate probability mass across near-equivalent facts.



Display abbreviates subjects; full overlap example uses “The research team”.

Similarity: 12/13 ≈ 0.923 ≥ 0.9 | Rank clusters; surface each most frequent member.

n = candidate-pool budget k = returned-cluster budget Reported: n = 512, k = 4

Paper §3.4, Eqs. (2–3) · Synthetic candidate counts shown only to explain aggregation

Uniform corruption wins in the four-tag regime.

REVISE ROLE ASSIGNMENTS



Uniform: tag ↔ tag

Absorbing: [MASK] → tag
Revision requires remasking.

DEVELOPMENT F1

CaRB

50.2 ± 0.2 ⟶ **52.4 ± 0.2** **+2.2**

CaRB (1–1)

49.7 ± 0.2 ⟶ **53.1 ± 0.2** **+3.4**

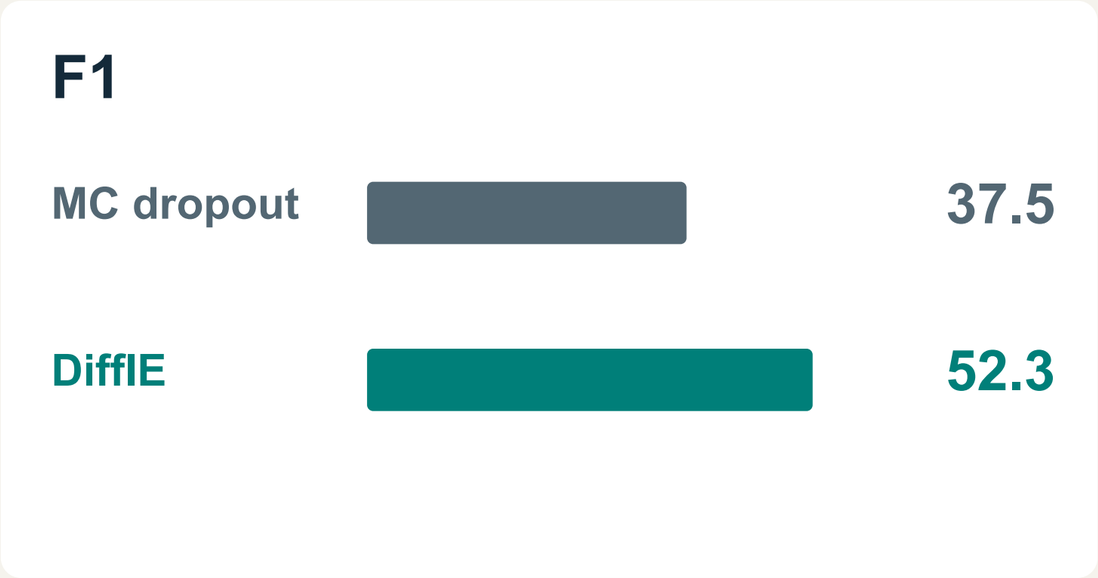
Tuned MDLM **Uniform D3PM**

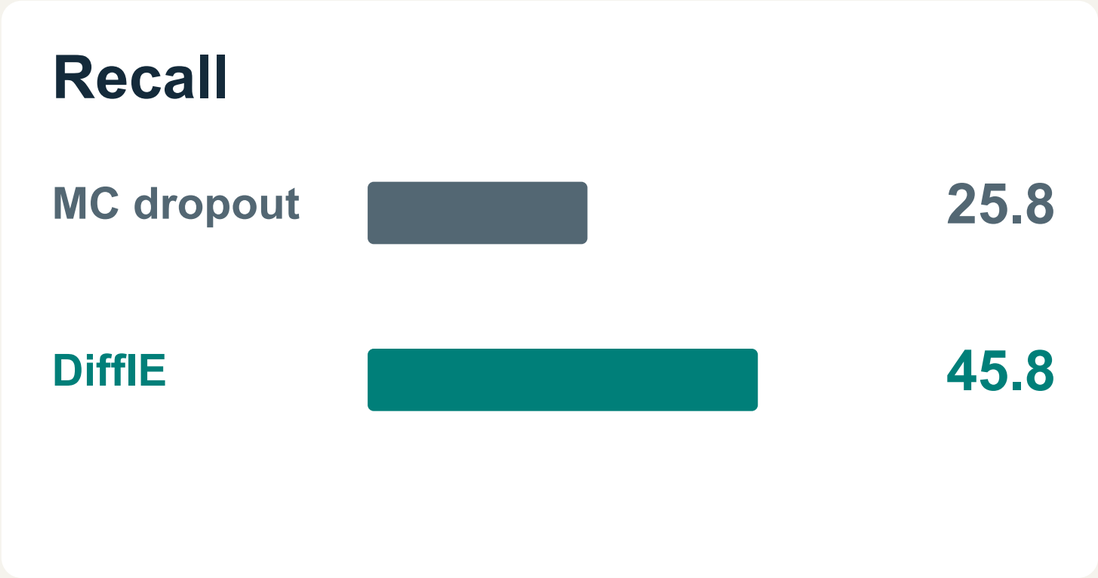
Evidence for this small-vocabulary setting; revisability is the proposed explanation.

Paper §6.3, Table 5 · CaRB development · 2.5K training sentences · Mean ± SD, 10 seeds

Repeated stochastic tagging does not recover the gains.

Shared encoder · training instances · labels · span construction · clustering





The gap is in candidate generation: aggregation cannot recover facts that are never proposed.

Choose the compute–quality operating point after training.

TRAJECTORIES n

16

49.3 F1

14.2 sentences / sec

TRAJECTORIES n

64

50.8 F1

6.3 sentences / sec

TRAJECTORIES n

512

52.3 F1

0.95 sentences / sec

16 reverse steps per trajectory · encoder states reused · n changed only at inference

512 → 64 trajectories: 6.6× throughput for −1.5 F1

Paper §6.2, Table 4 · CaRB development subset · End-to-end inference, single NVIDIA A100

The strongest gains appear under stricter matching.

CaRB (1–1) F1

51.9 ± 0.2

Previous reported best

51.5 · DualOIE

CaRB (1–1) AUC

34.5 ± 0.1

Previous reported best

33.6 · CycleOIE

BenchIE F1

34.3 ± 0.3

Previous reported best

34.0 · ClausIE

Standard CaRB F1: 52.2 vs 56.3 (DualOIE)

WiRe57 F1: 36.1 vs 40.0 (OpenIE6)

2,500 training sentences · 8,266 aligned triplet instances · one globally selected inference setting

Paper §5, Table 2; Appendix A, Tables 7–8 · Published baseline scores; DiffIE means over 10 seeds

Diffusion diversity can recover structured output sets.

01 **Model a distribution**

Discrete diffusion over four per-token roles.

02 **Use diversity**

Independent trajectories recover candidate facts.

03 **Control inference**

Tune the pool size n and output budget k after training.

Thank you · Questions

What exactly is learned and sampled?

TRAINING EXAMPLES

$(x, \text{triplet}_1), \ldots, (x, \text{triplet}_m)$
$\rightarrow$ m independently shuffled taggings

$$L = E\left[- \Sigma_i \, w_i \log p\theta(y_i^{(0)} \mid y^{(t)}, x, t) \right]$$

Background weight: 0.7 Subject / relation / object weights: 1.0

Inference: uniform random tags → 16 reverse steps → longest contiguous role spans

Clustering uses role-wise word overlap and cluster mass.

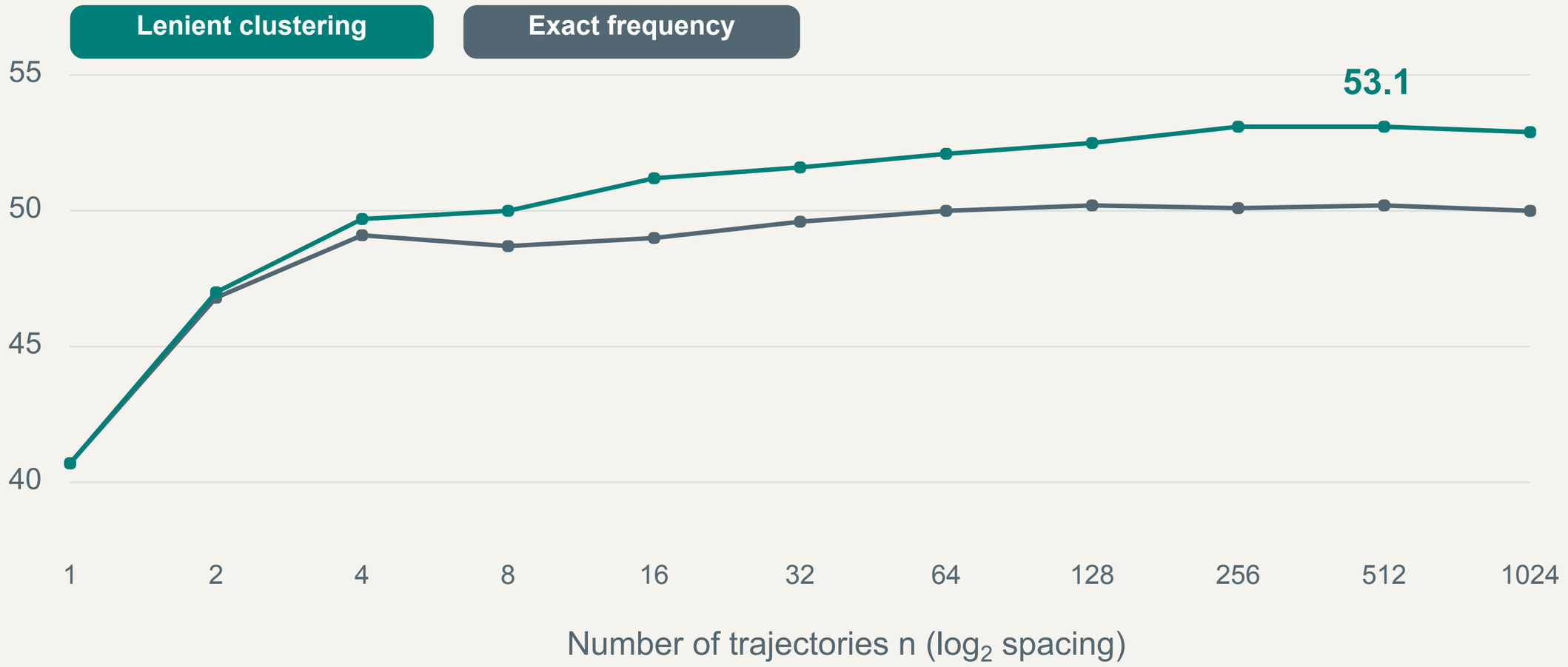
$$F_{ij} = 2m_{ij} / (n_i + n_j)$$

$$\text{score(cluster)} = \text{count} / n$$

1. Connect triplets with role-wise overlap F1 ≥ 0.9.
2. Take connected components; sum member frequencies.
3. Rank clusters by mass and return up to k clusters.
4. Surface the highest-frequency member of each cluster.

Paper §3.4, Eqs. (2–3) · τ = 0.9, k = 4, n = 512 selected on CaRB development

More trajectories show diminishing returns.



Paper Appendix B, Table 10 · CaRB (1–1) development F1; fixed τ and k

Full benchmark comparison

System	CaRB F1	CaRB AUC	1–1 F1	1–1 AUC	BenchIE F1	WiRe57 F1
OpenIE6	52.7	33.7	46.4	26.8	25.0	40.0
IMoJIE	53.5	33.3	41.4	22.2	18.6	36.0
DetIE	52.1	36.7	40.1	29.3	—	36.0
ClausIE	45.0	22.0	40.2	17.7	34.0	33.2
CompactIE	45.0	—	—	—	26.2	31.8
DualOIE	56.3	—	51.5	—	—	—
ChatGPT (CoT)	53.4	—	—	—	—	—
CycleOIE	51.2	39.0	47.4	33.6	—	—
DiffIE	**52.2**	**37.1**	**51.9**	**34.5**	**34.3**	**36.1**

Paper Table 2 · Scores in percentage points · — = not reported · DiffIE: mean over 10 seeds

Where the current method is limited

REPRESENTATION

One role per token

Longest-span construction can discard discontinuous arguments.

COMPUTATION

Sampling has a cost

Higher n increases runtime. Fixed-slot tagging remains much faster.

EVALUATION SCOPE

English, extractive data

Curated LSOIE-derived labels; no multilingual or out-of-domain evaluation.

Future direction: richer span construction and other tasks with multiple valid outputs.

Paper Limitations; §3.1, §6.2 · Discontinuous-run statistics measured on CaRB development

Reported training and inference configuration

Encoder	bert-base-uncased; lowest 4 layers frozen
Denoiser	6 layers · width 512 · 8 attention heads
Noise process	Uniform D3PM · cosine schedule · s = 0.002
Optimization	AdamW · 12 epochs · batch size 32
Learning rates	Denoiser 2×10^{-4} · encoder 5×10^{-5}
Regularization	Weight decay 0.01 · 500 warmup steps
Inference	$T = 16 \cdot n = 512 \cdot k = 4 \cdot \tau = 0.9$

Paper §4 · Model and inference hyperparameters selected on CaRB development